A Notebook on Graph Theory and Topology

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To my family, friends and communities members who have been dedicating to the presentation of this notebook, and to all students, researchers and faculty members who might find this notebook helpful.

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Foreword

I was not a big supporter for sharing notebooks. Notebooks, in my view, had been private and practically useless to anyone other than their owner.

It traces back to the summer of 2009 when I started my third year as a high school student. In the end of August when the results of Gaokao (National College Entrance Examination of China, annually held in July) came out, photocopy shops would start selling notebooks photocopies that they claimed to be from the top scorers of the exam. Much curious as I was about what these notebooks look like, never have I purchased any of them, and nor expected myself to actually learn anything from them. In fact, many of these notebooks were more difficult to read than the textbooks. I guess we could not blame the top scorers for being so smart that they sometimes made things extremely brief or overwhelmingly comprehensive. Not to mention, why would I want to adapt to notebooks of others when I had my own notebooks which in my opinion should be just as good as theirs.

However, my mind changed after becoming an undergraduate student in 2010 with the school of electrical engineering. There were so many engineering modules, and as an unfortunate result it was very challenging to dig deep into all of them evenly. I naturally put less effort on "side" modules such as material science than my major's core modules such as analog electronics. The situation became worse when I started pursuing my Ph.D. in 2014. As I had to focus on specific research areas entirely, I could hardly split enough time to systematically study those less-relevant yet interesting contents. For example, my research area was on robust statistics, state estimation and adaptive control, and I could not spend as much time as I wished on graph theory. I needed a way to quickly scratch the surface of some areas, and became decently familiar with them. There was no intention to become an expert on these areas by all means. I just wanted to understand what they are, and see whether they could have inspired my own research.

To make a difference, I forced myself to read articles beyond my comfort zone and it ended up motivating me to take notes to consolidate the knowledge. I started with easy subjects. My very first notebook was on Numerical Analysis, an entrance-level module for engineering background graduate students. Revisiting that module did not bring much benefit to my research, but it served as a good starting point. Following that were some more math notebooks, then a few on computing, operating systems and databases, and then on data science, and then on ontology.

Eventually at one point, it suddenly came to me: these notebooks that I had taken while learning these subjects as an armature, they were exactly the materials I needed, quoting myself, to "quickly scratch the surface of some areas, and became decently familiar with them". Well, in that case, why not digitizing them, making them accessible online and open-source and letting everyone read and edit it?

Hence, this is where we are.

Similar with most open-source software, these notebooks do not come with any "warranty" of any kind, meaning that there is no guarantee that everything in the notebooks is correct. Always bear in mind that the notebooks are not peer reviewed. **Do NOT cite this notebook or any notebooks in this series in your academic research paper!** If you find anything helpful here and you want to use it in your research, please trace back to the origin of the knowledge and confirm by yourself.

This notebook is suitable as:

- a quick reference guide;
- a brief introduction for beginners of an area;
- a "cheat sheet" for students or teachers for an entrance-level course.

This notebook is NOT suitable as:

- a direct research reference;
- a replacement of the corresponding textbook.

Again, this notebook is NOT peer reviewed. It is meant to be easy to read, not to be comprehensive and very rigorous.

Although this notebook is open-source, the reference materials of this notebook, including textbooks, journal papers, conference proceedings, etc., may not be open-source. Very likely many of these reference materials are licensed or copyrighted. Please legitimately access these materials and properly use them, should you decided to trace the origin of the knowledge.

Some of the figures in this notebook are plotted using Excalidraw, a very convenient tool to emulate hand drawings. The Excalidraw project can be found on GitHub, *excalidraw/excalidraw*. Other figures may come from MATLAB, R, Python, and other computation engines. The source code to reproduce the results are likely to be included in the same repository of the notebook, but there might be exceptions.

This work might have benefited from the assistance of large language models, which are used exclusively for editing purposes such as correcting grammar and rephrasing sentences, without introducing new content, generating innovative information, or changing the original intent of the text.

Preface

This notebook discusses graph theory and general topology.

A list of key references of this notebook is given below. These materials are frequently cited in the notebook. For convenience, they are not given in the Reference section but listed here instead, whereas otherwise I will have to cite them everywhere in the text.

Topology

- Willard, Stephen. General topology. Courier Corporation, 2012.
- MIT Open Course, Introduction To Metric Spaces

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Graph Theory

Part II.

General Topology

1. Set Theory and Metric Space

Set theory and metric space are introduced in this chapter as the preliminaries to general topology. They are also the building blocks of many other mathematics subjects such as algebra and geometry.

1.1. Set

TBA

1.2. Metric Space

Metric space and relevant concepts are introduced in this section. Commonly used metric space and their use cases are briefly covered.

1.2.1. Definition

A **metric space** is a non-empty set X with a distance function $d : X \times X \to \mathbb{R}$, where function d satisfies the following properties:

- Symmetric: $\forall x, y \in X,\ d(x, y) = d(y, x)$;
- Positive definite: $\forall x, y \in X,\ d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$;
- Triangle inequality: $\forall x, y, z \in X,\ d(x, z) \geq d(x, y) + d(y, z)$.

Is metric space an algebraic system?

No, we do not consider metric space an algebraic system. The distance function defined in a metric space is essentially a mapping but not an operator. The tools and properties we study on metric spaces and algebraic systems are also different.

In that perspective, though both metric space and algebraic system define set and "rules" of some kind, they are different.

Consider the following metric space (X, d), where the distance function is defined by

$$d(x, y) = \begin{cases} 0 & x = y \\ 1 & \text{otherwise} \end{cases} \tag{1.1}$$

It can be easily proved that the above meets the criteria of metric space, hence (X, d) a metric space. This is probably one of the simplest metric spaces and it demonstrates

that metric space can be defined on any non-empty set. But of course, this metric space is not very interesting, nor useful.

In the next Sections 1.2.2, 1.2.3 and 1.2.4, commonly seen metric spaces and concepts defined using these metric spaces are introduced. These metric spaces and concepts are widely used in many mathematics subjects including algebra and topology.

1.2.2. Open Set

In the context of metric space, define **ball of radius** r **centered at** p as follows.

$$B_r(p) \quad = \quad \{x \in X | d(x,p) < r\}$$

With that defined, open set can be defined as follows. A set $A \subseteq X$ is an **open set** if $\forall a \in A$, $\exists \epsilon > 0$ such that $B_\epsilon(a) \subseteq A$.

Open set is an important concept defined on top of metric space and it is widely used in the remainder of this notebook.

1.2.3. Vector and Relevant Metric Spaces

The definition of vector space is formally introduced in linear algebra relevant notebooks. A brief is given here. A **vector space** refers to an algebraic system defined on a set V (the elements in V are known as vectors) with the following operators

$$\begin{aligned} \oplus &: V \times V \to V, \\ \odot &: \mathbb{R} \times V \to V, \end{aligned}$$

where $\oplus$ is the addition operator that combines two vectors to make a third vector inside the set, and $\odot$ the scalar multiplication operator that scale the vector by a scalar to get a new vector inside the set. Notice that a vector space is always closed under both addition and scalar multiplication.

Commonly seen vector spaces include $\mathbb{R}^n$ which is represented by $[x_1, \ldots, x_n]^T$. The $\oplus$ and $\odot$ operators are defined by

$$\begin{aligned} x + y &= \begin{bmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{bmatrix}, \\ cx &= \begin{bmatrix} cx_1 \\ \vdots \\ cx_n \end{bmatrix}, \end{aligned}$$

respectively.

Euclidean distance (also known as l^2**-distance**) is a metric space defined on $\mathbb{R}^n$, with the distance function defined by

$$\|x - y\|_2 \quad = \quad \left(\sum_n |x_i - y_i|^2 \right)^{\frac{1}{2}} \tag{1.2}$$

where $x, y \in \mathbb{R}^n$ with i the element index of the vector. Euclidean distance is one of the most popular metric spaces in science and engineering, and it will be frequently used in the remainder of this notebook,

Another example of metric space is the l^∞**-distance** on $\mathbb{R}^n$ with the distance function defined by

$$\|x - y\|_\infty = \max_n |x_i - y_i|. \tag{1.3}$$

It can be easily verified that both Euclidean distance and l^∞-distance on $\mathbb{R}^n$ satisfy the definition of metric space.

Both Euclidean distance and l^∞-distance can be derived from p**-norm** which is defined on $\mathbb{R}^n$ as follows.

$$\|x\|_p = \left(\sum_n |x_i|^p\right)^{\frac{1}{p}} \tag{1.4}$$

where $p > 0$. Under the scope of this notebook, only positive integer p is considered.

Letting $p = 2$ in (1.4) gives

$$\|x\|_2 = \left(\sum_n x_i^2\right)^{\frac{1}{2}}$$

which is known as the **Euclidean norm**, and Euclidean distance in (1.2) is the Euclidean norm of $x - y$.

Letting $p \to \infty$ in (1.4) gives

$$\|x\|_\infty = \max_n |x_i|$$

which is known as the **infinity norm**, and l^∞-distance in (1.2) is the infinity norm of $x - y$.

Likewise, p-norm with other values of p can be used as distance functions and a series of metric spaces can be defined. For example, (1.4) corresponding with $p = 1$, i.e. 1-norm $\|\cdot\|_1$, can be used to define **rectilinear distance** (also known as **Manhattan distance**) $\|x - y\|_1$.

A handful property about p-norm is given below.

$$\|x\|_{p_1} \leq \|x\|_{p_2}, \text{if } p_1 \geq p_2.$$

So far we have been discussing $x \in \mathbb{R}^n$ which has finite elements. Consider $\{x\}$ a sequence with infinite elements. In this case, the p-norm of an infinite sequence can be defined as follows

$$\|x\|_p = \left(\sum_{i=1}^{\infty} |x_i|^p\right)^{\frac{1}{p}} \tag{1.5}$$

which may be bounded or unbounded. Define l^p **space** which is the collection of infinite sequence whose corresponding (1.5) is bounded. For instance, l^1 space refers to the space of infinite sequences whose values are absolutely convergent, and l^2 space refers to the space of infinite sequences whose values are square-summable.

1.2.4. Continuous Function and Relevant Metric Spaces

There are different ways to define a continuous function. In the essence, continuous function refers to a function where if the inputs are "very close", their corresponding outputs must also be very close, and can be made arbitrarily close. From that perspective, the ϵ-δ definition which uses metric space can be used to define continuous function as follows.

Let (X, d_x), (Y, d_y) be two metric spaces defined on sets X and Y respectively, and let $f : X \to Y$ be a function mapping elements from X to Y. Function f is a **continuous function** at x_0 if $\forall \epsilon > 0$, $\exists \delta > 0$ such that

$$d(x, x_0) < \delta \quad \Longrightarrow \quad d\left(f(x), f(x_0)\right) < \epsilon$$

Let $f, g \in C^0\left([a, b]\right)$ be two continuous functions on interval $[a, b] \subset \mathbb{R}$, and $f, g : [a, b] \to \mathbb{R}$. Define the distance function

$$d(f, g) \quad = \quad \sup_{x \in [a,b]} |f(x) - g(x)|.$$

It can be proved that $\left(C^0\left([a, b]\right), d\right)$ is a metric space. This is demonstrated by Fig. 1.1, where the red and blue curve are two continuous functions and their distance on $[a, b]$ is given by the black solid segment.

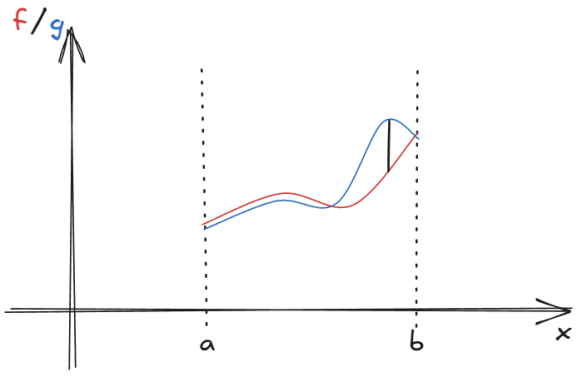


Figure 1.1.: A demonstrative example of a metric space defined on continuous functions.

The proof of $\left(C^0\left([a, b]\right), d\right)$ to be a metric space can be done fairly easily using the definition, and it is neglected here. The same idea applies to multivariable functions.

A function is a **continuously differentiable function** if it is differentiable in the given interval, and its differential is continuous in the interval. A continuously differentiable function is often denoted by $C^1[a, b]$. Define distance function

$$d(f, g) \quad = \quad \sup_{x \in [a,b]} |f(x) - g(x)| + \sup_{x \in [a,b]} |f'(x) - g'(x)|$$

and $\left(C^1[a,b],d\right)$ is a metric space.

The idea extends to **n-th order continuously differentiable function** which refers to a function that is at least n-th order differentiable in the interval with all these differentials continuous functions. An n-th order continuously differentiable function is often denoted by $C^n[a,b]$. Define distance function

$$d(f,g) = \sum_{i=0}^{n} \sup_{x\in[a,b]} |f^{(i)}(x) - g^{(i)}(x)| \tag{1.6}$$

and $(C^n[a,b],d)$ is a metric space.

An **infinity-order continuously differentiable function**, also known as a **smooth function**, is a function that has any order of differentiable, all of which continuous in the interval. It is often denoted by $C^\infty[a,b]$. Consider reusing (1.6) with $n\to\infty$ as the distance function on smooth functions to form a metric space. A practical issue is that (1.6) might be unbounded. We have to either limit the set of the metric space to only those smooth functions with finite (1.6), or to use a different distance function such as

$$d(f,g) = \sum_{i=0}^{\infty} 2^{-i} \frac{\sup_{x\in[a,b]}|f^{(i)}(x) - g^{(i)}(x)|}{1+\sup_{x\in[a,b]}|f^{(i)}(x) - g^{(i)}(x)|}.$$

Consider another metric space defined on continuous functions.

Let $f,g\in C^0\left([a,b]\right)$. Define the distance function $I_1 : f\times g\to\mathbb{R}$ as follows.

$$I_1(f,g) = \int_a^b |f(x)-g(x)|dx.$$

It can be proved that the above forms a metric space.

The idea can be extended to the following series of distance functions, each corresponding with a metric space.

$$I_p(f,g) = \left(\int_a^b |f(x)-g(x)|^p dx\right)^{\frac{1}{p}}, \tag{1.7}$$

where $p\geq 1$. Under the scope of this notebook, only integer p is considered. It can be proved that each different choice of p corresponds with a metric space.

Inspired by (1.7), define the L^p **norm** of a continuous function as follows

$$\|f\|_p = \left(\int_a^b |f(x)|^p dx\right)^{\frac{1}{p}}$$

The L^p norm of a continuous function may be bounded or unbounded. Continuous functions with bounded L^p norm form the L^p **space**.

Notice that L^p norm and L^p space are defined on the set of continuous functions, whereas l^p norm and l^p space introduced in earlier Section 1.2.3 are defined on the set of vectors and sequences, and hence they are different. That said, there are clear relations on the way their distance functions are defined.

Bibliography